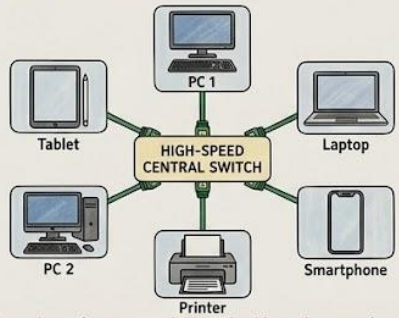
NETWORK TOPOLOGY

Q1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

Ans:-

1. STAR TOPOLOGY



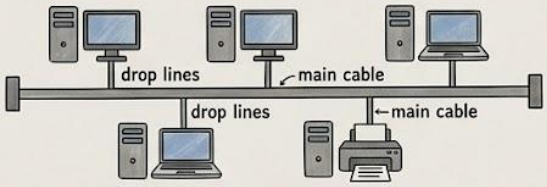
Every device has its own dedicated cable to the central switch.



Real-world app/service: Flipkart and College Campuses.

Star topology allows countless shoppers or students to connect to central servers; a failure of one connection (e.g., in a student hostel or individual app user) does not affect others.

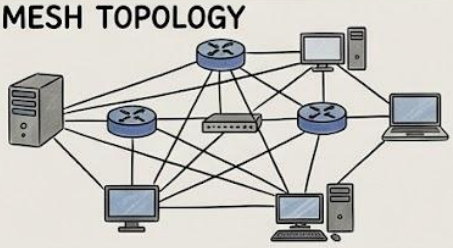
2. BUS TOPOLOGY



Real-world app/service: Older computer labs or home networks.

Bus topology connects devices via a single shared cable, which is simple and cost-effective for small groups, though a break in the main cable takes down the entire network.

3. MESH TOPOLOGY



Real-world app/service: WhatsApp, Public Safety networks, and Data Centers.

Mesh topology offers extreme redundancy; if one path or device fails (e.g., a city-wide power outage), data can be rerouted through multiple other links, ensuring continuous service for messaging or critical communications.

Q2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Ans:-

Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case (with example)
Star	• Easy to install and manage • Failure of one device does not affect others • Easy to find and fix faults • Easy to add new devices	• If the central switch fails, the whole network stops • Needs more cable than bus • Central switch adds cost • Speed depends on the switch capacity	Home, college or office WiFi/LAN. Example: your phone connecting to a WiFi router to use Instagram or Zomato.
Bus	• Cheap, needs less cable • Simple to set up • Good for very small networks • Easy to extend a little for temporary setups	• If the main cable breaks, the whole network fails • Slows down as more devices are added • Hard to troubleshoot • Limited cable length and number of devices	Small temporary LAN, e.g. a classroom computer lab. Not used by big apps like Instagram or Paytm.
Ring	• Every device gets equal chance to send data • Performance stays predictable under load • No central device needed • Fairly easy to install	• One broken link can stop the network (single ring) • Adding or removing a device disturbs the network • Troubleshooting is harder • Data passes through every node, so it gets slower with more devices	Telecom and ISP fiber rings connecting cities. Example: the fiber ring that carries Instagram and Paytm traffic between cities.
Topology	**Pros (Advantages)**	**Cons (Disadvantages)**	**Best Use Case (with example)**
Mesh	• Very reliable, many alternate paths • Fast, no traffic bottleneck • One failed link does not stop others • Secure, private point-to-point links	• Very expensive (many cables and ports) • Complex to set up • Difficult to maintain • Many redundant connections	Internet backbone, data centers and mesh WiFi. Example: Paytm/UPI payments, which need very high reliability.

Q3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

Ans:-

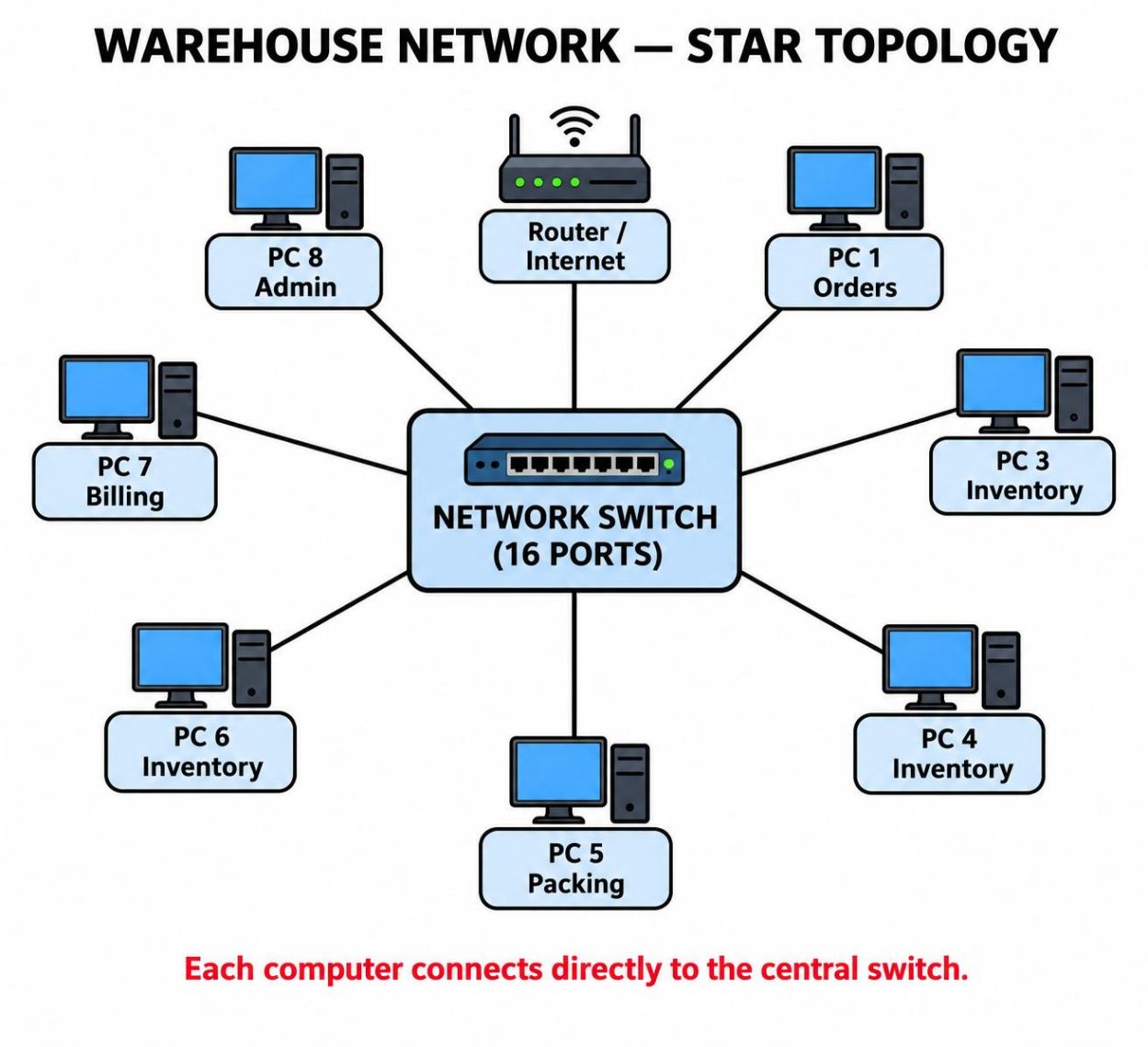
Chosen topology: Hybrid of Star and Mesh. Players connect in a star to their nearest game server, and the game servers are linked to each other in a mesh.

- **Speed:** Players connect to a nearby server, so ping (delay) is low, and servers linked directly share match data quickly.
- **Reliability:** If one server fails, players can be moved to another server through the mesh links, so the game keeps running.
- **Cost:** A full mesh for every player would be too costly, so mesh is used only between a few servers and star for players. This keeps the cost moderate.

Q4. Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Ans:-

Recommended topology: Star Topology



* **Reasoning:** A star network needs only one affordable switch and simple cables, so the cost is low. If one computer or cable has a problem, only that computer is affected and the fault is easy to find. For future expansion, a new computer can simply be plugged into a free port on the switch (a 16-port switch leaves room for 8 more).